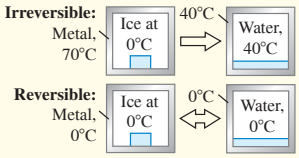
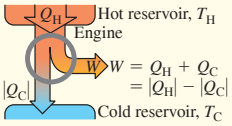
CHAPTER 20 SUMMARY

Reversible and irreversible processes: A reversible process is one whose direction can be reversed by an infinitesimal change in the conditions of the process, and in which the system is always in or very close to thermal equilibrium. All other thermodynamic processes are irreversible.



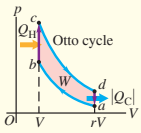
Heat engines: A heat engine takes heat Q_H from a source, converts part of it to work W, and discards the remainder $|Q_C|$ at a lower temperature. The thermal efficiency e of a heat engine measures how much of the absorbed heat is converted to work. (See Example 20.1.)

$$e = \frac{W}{Q_H} = 1 + \frac{Q_C}{Q_H} = 1 - \left|\frac{Q_C}{Q_H}\right| \qquad (20.4)$$



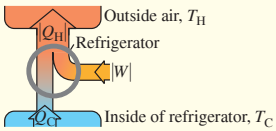
The Otto cycle: A petrol engine operating on the Otto cycle has a theoretical maximum thermal efficiency e that depends on the compression ratio r and the ratio of heat capacities γ of the working substance.

$$e = 1 - \frac{1}{r^{\gamma-1}} \qquad (20.6)$$

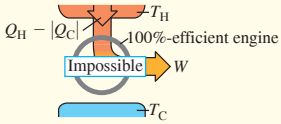


Refrigerators: A refrigerator takes heat Q_C from a colder place, has a work input $|W|$, and discards heat $|Q_H|$ at a warmer place. The effectiveness of the refrigerator is given by its coefficient of performance K.

$$K = \frac{|Q_C|}{|W|} = \frac{|Q_C|}{|Q_H| - |Q_C|} \qquad (20.9)$$

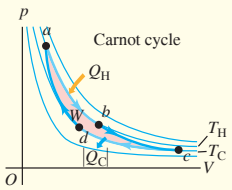


The second law of thermodynamics: The second law of thermodynamics describes the directionality of natural thermodynamic processes. It can be stated in two equivalent forms. The *engine* statement is that no cyclic process can convert heat completely into work. The *refrigerator* statement is that no cyclic process can transfer heat from a colder place to a hotter place with no input of mechanical work.



The Carnot cycle: The Carnot cycle operates between two heat reservoirs at temperatures T_H and T_C and uses only reversible processes. Its thermal efficiency depends only on T_H and T_C. An additional equivalent statement of the second law is that no engine operating between the same two temperatures can be more efficient than a Carnot engine. (See Examples 20.2 and 20.3.)

$$e_{Carnot} = 1 - \frac{T_C}{T_H} = \frac{T_H - T_C}{T_H} \qquad (20.14)$$



A Carnot engine run backward is a Carnot refrigerator. Its coefficient of performance depends on only T_H and T_C. Another form of the second law states that no refrigerator operating between the same two temperatures can have a larger coefficient of performance than a Carnot refrigerator. (See Example 20.4.)

$$K_{Carnot} = \frac{T_C}{T_H - T_C} \qquad (20.15)$$

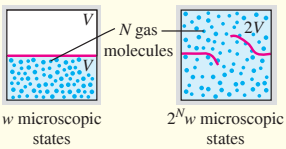
Entropy: Entropy is a quantitative measure of the randomness of a system. The entropy change in any reversible process depends on the amount of heat flow and the absolute temperature T. Entropy depends only on the state of the system, and the change in entropy between given initial and final states is the same for all processes leading from one state to the other. This fact can be used to find the entropy change in an irreversible process. (See Examples 20.5–20.10.)

An important statement of the second law of thermodynamics is that the entropy of an isolated system may increase but can never decrease. When a system interacts with its surroundings, the total entropy change of system and surroundings can never decrease. When the interaction involves only reversible processes, the total entropy is constant and $\Delta S = 0$; when there is any irreversible process, the total entropy increases and $\Delta S > 0$.

$$\Delta S = \int_1^2 \frac{dQ}{T} \quad \text{(reversible process)} \tag{20.19}$$

Entropy and microscopic states: When a system is in a particular macroscopic state, the particles that make up the system may be in any of w possible microscopic states. The greater the number w, the greater the entropy. (See Example 20.11.)

$$S = k \ln w \tag{20.22}$$



Chapter 20 Media Assets

For assigned homework and other learning materials, go to Mastering Physics.

GUIDED PRACTICE

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLE 20.1 (Section 20.2) before attempting these problems.

VP20.1.1 A diesel engine has efficiency 0.180. (a) In order for this engine to do 1.24×10^4 J of work, how many joules of heat must it take in? (b) How many joules of this heat is discarded?

VP20.1.2 In one cycle a heat engine absorbs 3.82×10^4 J of heat from the hot reservoir and rejects 3.16×10^4 J of heat to the cold reservoir. What is the efficiency of this engine?

VP20.1.3 Measurements of a petrol engine show that it has an efficiency of 0.196 and that it exhausts 4.96×10^8 J of heat during 20 minutes of operation. During that time, (a) how much heat does the engine take in and (b) how much work does the engine do?

VP20.1.4 An aircraft piston engine that burns petrol (heat of combustion 5.0×10^7 J/kg) has a power output of 1.10×10^5 W. (a) How much work does this engine do in 1.00 h? (b) This engine burns 34 kg of petrol per hour. How much heat does the engine take in per hour? (c) What is the efficiency of the engine?

Be sure to review EXAMPLES 20.2, 20.3 and 20.4 (Section 20.6) before attempting these problems.

VP20.4.1 In one cycle a Carnot engine takes in 8.00×10^4 J of heat and does 1.68×10^4 J of work. The temperature of the engine's cold reservoir is 25.0°C. (a) What is the efficiency of this engine? (b) How much heat does this engine exhaust per cycle? (c) What is the temperature (in °C) of the hot reservoir?

VP20.4.2 For the Carnot cycle described in Example 20.3, you change the temperature of the cold reservoir from 27°C to −73°C. The initial pressure and volume at point a are unchanged, the volume still doubles during the isothermal expansion $a \rightarrow b$, and the volume still decreases by one-half during the isothermal compression $c \rightarrow d$. For this modified cycle, calculate (a) the new efficiency of the cycle and (b) the amount of work done in each of the four steps of the cycle.

VP20.4.3 A Carnot refrigerator has a cold reservoir at −10.0°C and a hot reservoir at 25.0°C. (a) What is its coefficient of performance? (b) How much work input does this refrigerator require to remove 4.00×10^6 J of heat from the cold reservoir?

VP20.4.4 A Carnot engine uses the expansion and compression of n moles of argon gas, for which $C_V = \frac{3}{2}R$. This engine operates between temperatures T_C and T_H. During the isothermal expansion $a \rightarrow b$, the volume of the gas increases from V_a to $V_b = 2V_a$. (a) Calculate the work W_{ab} done during the isothermal expansion $a \rightarrow b$. Give your answer in terms of n, R, and T_H. (b) Calculate the work W_{bc} done during the adiabatic expansion $b \rightarrow c$. Give your answer in terms of n, R, T_C and T_H. (c) For this engine, $W_{ab} = W_{bc}$. Find the ratio T_C/T_H and the efficiency of the engine.

Be sure to review EXAMPLES 20.6, 20.7, 20.8, 20.9, and 20.10 (Section 20.7) before attempting these problems.

VP20.10.1 Ethanol melts at 159 K (heat of fusion 1.042×10^5 J/kg) and boils at 351 K (heat of vaporization 8.54×10^5 J/kg). Liquid ethanol has a specific heat of 2428 J/kg · K (which we assume does not depend on temperature). If you have 1.00 kg of ethanol originally in the solid state at 159 K, calculate the change in entropy of the ethanol when it (a) melts at 159 K, (b) increases in temperature as a liquid from 159 K to 351 K, and (c) boils at 351 K.

VP20.10.2 Initially 5.00 mol of helium (which we can treat as an ideal gas) occupies volume 0.120 m^3 and is at temperature 20.0°C. You allow

the helium to expand so that its final volume is 0.360 m^3 and its final temperature is also 20.0°C. Calculate the net change in entropy of the helium if (a) you make the helium expand isothermally from 0.120 m^3 to 0.360 m^3 and (b) you first increase the temperature of the helium so that it expands at constant pressure from 0.120 m^3 to 0.360 m^3, then cool the helium at constant volume to 20.0°C.

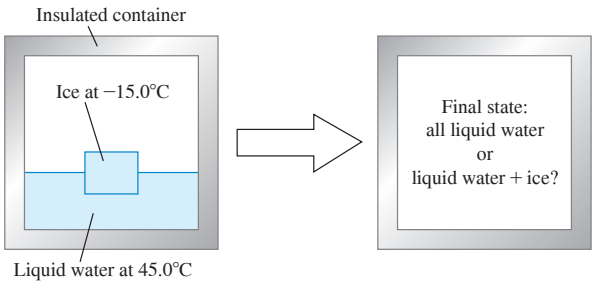
VP20.10.3 In one cycle of operation, an engine that does *not* use the Carnot cycle takes in 8.00×10^4 J of heat from a reservoir at 260°C, does 1.60×10^4 J of work, and rejects 6.40×10^4 J of heat to a reservoir at 20°C. At the end of the cycle, the engine is in the same state as at the beginning of the cycle. Calculate the change in entropy in one cycle for (a) the engine, (b) the reservoir at 260°C, (c) the reservoir at 20°C, and (d) the system of engine and two reservoirs. Does the net entropy of the system increase, decrease, or stay the same?

VP20.10.4 You combine 1.00 kg of ice (heat of fusion 3.34×10^5 J/K) at 0.0°C and 0.839 kg of liquid water (specific heat 4.19×10^3 J/K)) at 95.0°C. When the system comes to equilibrium, all of the ice has melted and the temperature of the mixture is 0.0°C. Calculate the change in entropy for (a) the ice as it melts, (b) the water that was initially at 95.0°C, and (c) the combination of the two. Does the net entropy of the system increase, decrease, or stay the same?

BRIDGING PROBLEM Entropy Changes: Cold Ice in Hot Water

An insulated container of negligible mass holds 0.600 kg of water at 45.0°C. You put a 0.0500 kg ice cube at −15.0°C in the water (**Fig. 20.23**). (a) Calculate the final temperature of the water once the ice has melted. (b) Calculate the change in entropy of the system.

Figure **20.23** What becomes of this ice–water mixture?



SOLUTION GUIDE

IDENTIFY and SET UP

1. Make a list of the known and unknown quantities, and identify the target variables.
2. How will you find the final temperature of the ice–water mixture? How will you decide whether or not all the ice melts?
3. Once you find the final temperature of the mixture, how will you determine the changes in entropy of (i) the ice initially at −15.0°C and (ii) the water initially at 45.0°C?

EXECUTE

4. Use the methods of Chapter 17 to calculate the final temperature *T*. (*Hint:* First assume that all of the ice melts, then write an equation which says that the heat that flows into the ice equals the heat that flows out of the water. If your assumption is correct, the final temperature that you calculate will be greater than 0°C. If your assumption is incorrect, the final temperature will be 0°C or less, which means that some ice remains. You'll then need to redo the calculation to account for this.)
5. Use your result from step 4 to calculate the entropy changes of the ice and the water. (*Hint:* You must include the heat flow associated with temperature changes, as in Example 20.6, as well as the heat flow associated with the change of phase.)
6. Find the total change in entropy of the system.

EVALUATE

7. Do the signs of the entropy changes make sense? Why or why not?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q20.1 A pot is half-filled with water, and a lid is placed on it, forming a tight seal so that no water vapor can escape. The pot is heated on a stove, forming water vapor inside the pot. The heat is then turned off and the water vapor condenses back to liquid. Is this cycle reversible or irreversible? Why?

Q20.2 Give two examples of reversible processes and two examples of irreversible processes in purely mechanical systems, such as blocks sliding on planes, springs, pulleys, and strings. Explain what makes each process reversible or irreversible.

Q20.3 Household refrigerators have arrays or coils of tubing on the outside, usually at the back or bottom. When the refrigerator is running, the tubing becomes quite hot. Where does the heat come from?

Q20.4 Suppose you try to cool the kitchen of your house by leaving the refrigerator door open. What happens? Why? Would the result be the same if you left open a picnic cooler full of ice? Explain the reason for any differences.

Q20.5 Why must a room air conditioner be placed in a window rather than just set on the floor and plugged in? Why can a refrigerator be set on the floor and plugged in?

Q20.6 Is it a violation of the second law of thermodynamics to convert mechanical energy completely into heat? To convert heat completely into work? Explain your answers.

Q20.7 Imagine a special air filter placed in a window of a house. The tiny holes in the filter allow only air molecules moving faster than a certain speed to exit the house, and allow only air molecules moving slower than that speed to enter the house from outside. Explain why such an air filter would cool the house, and why the second law of thermodynamics makes building such a filter an impossible task.

Q20.8 An electric motor has its shaft coupled to that of an electric generator. The motor drives the generator, and some current from the generator is used to run the motor. The excess current is used to light a home. What is wrong with this scheme?

Q20.9 When a wet cloth is hung up in a hot wind in the desert, it is cooled by evaporation to a temperature that may be 20°C or so below that of the air. Discuss this process in light of the second law of thermodynamics.

Q20.10 Compare the pV-diagram for the Otto cycle in Fig. 20.6 with the diagram for the Carnot heat engine in Fig. 20.13. Explain some of the important differences between the two cycles.

Q20.11 The efficiency of heat engines is high when the temperature difference between the hot and cold reservoirs is large. Refrigerators, on the other hand, work better when the temperature difference is small. Thinking of the mechanical refrigeration cycle shown in Fig. 20.9, explain in physical terms why it takes less work to remove heat from the working substance if the two reservoirs (the inside of the refrigerator and the outside air) are at nearly the same temperature, than if the outside air is much warmer than the interior of the refrigerator.

Q20.12 What would be the efficiency of a Carnot engine operating with $T_H = T_C$? What would be the efficiency if $T_C = 0$ K and T_H were any temperature above 0 K? Interpret your answers.

Q20.13 Real heat engines, like the petrol engine in a car, always have some friction between their moving parts, although lubricants keep the friction to a minimum. Would a heat engine with completely frictionless parts be 100% efficient? Why or why not? Does the answer depend on whether or not the engine runs on the Carnot cycle? Again, why or why not?

Q20.14 Does a refrigerator full of food consume more power if the room temperature is 20°C than if it is 15°C? Or is the power consumption the same? Explain your reasoning.

Q20.15 In Example 20.4, a Carnot refrigerator requires a work input of only 230 J to extract 346 J of heat from the cold reservoir. Doesn't this discrepancy imply a violation of the law of conservation of energy? Explain why or why not.

Q20.16 How can the thermal conduction of heat from a hot object to a cold object increase entropy when the same amount of heat that flows out of the hot object flows into the cold one?

Q20.17 Explain why each of the following processes is an example of increasing randomness: mixing hot and cold water; free expansion of a gas; irreversible heat flow; developing heat by mechanical friction. Are entropy increases involved in all of these? Why or why not?

Q20.18 The free expansion of an ideal gas is an adiabatic process and so no heat is transferred. No work is done, so the internal energy does not change. Thus, $Q/T = 0$, yet the randomness of the system and thus its entropy have increased after the expansion. Why does Eq. (20.19) not apply to this situation?

Q20.19 Are the earth and sun in thermal equilibrium? Are there entropy changes associated with the transmission of energy from the sun to the earth? Does radiation differ from other modes of heat transfer with respect to entropy changes? Explain your reasoning.

Q20.20 Suppose that you put a hot object in thermal contact with a cold object and observe (much to your surprise) that heat flows from the cold object to the hot object, making the cold one colder and the hot one hotter. Does this process necessarily violate the first law of thermodynamics? The second law of thermodynamics? Explain.

Q20.21 If you run a movie film backward, it is as if the direction of time were reversed. In the time-reversed movie, would you see processes that violate conservation of energy? Conservation of linear momentum? Would you see processes that violate the second law of thermodynamics? In each case, if law-breaking processes could occur, give some examples.

Q20.22 **BIO** Some critics of biological evolution claim that it violates the second law of thermodynamics, since evolution involves simple life forms developing into more complex and more highly ordered organisms. Explain why this is not a valid argument against evolution.

Q20.23 **BIO** A growing plant creates a highly complex and organized structure out of simple materials such as air, water, and trace minerals. Does this violate the second law of thermodynamics? Why or why not? What is the plant's ultimate source of energy? Explain.

EXERCISES

Section 20.2 Heat Engines

20.1 • A diesel engine performs 2400 J of mechanical work and discards 4100 J of heat each cycle. (a) How much heat must be supplied to the engine in each cycle? (b) What is the thermal efficiency of the engine?

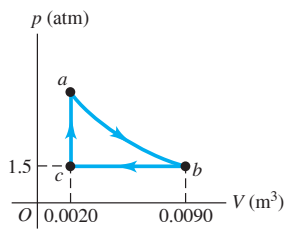
20.2 • An aircraft engine takes in 9000 J of heat and discards 6400 J each cycle. (a) What is the mechanical work output of the engine during one cycle? (b) What is the thermal efficiency of the engine?

20.3 • **A Petrol Engine.** A petrol engine takes in 1.61×10^4 J of heat and delivers 3700 J of work per cycle. The heat is obtained by burning petrol with a heat of combustion of 4.60×10^4 J/g. (a) What is the thermal efficiency? (b) How much heat is discarded in each cycle? (c) What mass of fuel is burned in each cycle? (d) If the engine goes through 60.0 cycles per second, what is its power output?

20.4 • A petrol engine has a power output of 200 kW. Its thermal efficiency is 25.0%. (a) How much heat must be supplied to the engine per second? (b) How much heat is discarded by the engine per second?

20.5 •• The pV-diagram in **Fig. E20.5** shows a cycle of a heat engine that uses 0.250 mol of an ideal gas with $\gamma = 1.40$. Process ab is adiabatic. (a) Find the pressure of the gas at point a. (b) How much heat enters this gas per cycle, and where does it happen? (c) How much heat leaves this gas in a cycle, and where does it occur? (d) How much work does this engine do in a cycle? (e) What is the thermal efficiency of the engine?

Figure **E20.5**



20.6 •• **CP** A heat engine uses a large insulated tank of ice water as its cold reservoir. In 100 cycles the engine takes in 8000 J of heat energy from the hot reservoir and the rejected heat melts 0.0180 kg of ice in the tank. During these 100 cycles, how much work is performed by the engine?

Section 20.3 Internal-Combustion Engines

20.7 • The theoretical efficiency of an Otto cycle is 63%. The ratio of the C_p and C_V heat capacities for the gas in the engine is $\frac{7}{5}$. (a) What is the compression ratio r? (b) If the net work done per cycle is 12.6 kJ, how much heat is rejected and how much is absorbed by the engine during each cycle?

20.8 • (a) Calculate the theoretical efficiency for an Otto-cycle engine with $\gamma = 1.40$ and $r = 9.50$. (b) If this engine takes in 10,000 J of heat from burning its fuel, how much heat does it discard to the outside air?

20.9 •• The Otto-cycle engine in a Porsche 911 Carrera has a compression ratio of 10.2. (a) What is the ideal efficiency of the engine? Use $\gamma = 1.40$. (b) The engine in a Ferrari F430 has a higher compression ratio of 11.3. How much increase in the ideal efficiency results from this increase in the compression ratio?

Section 20.4 Refrigerators

20.10 •• A freezer has a coefficient of performance of 2.40. The freezer is to convert 1.80 kg of water at 25.0°C to 1.80 kg of ice at −5.0°C in one hour. (a) What amount of heat must be removed from the water at 25.0°C to convert it to ice at −5.0°C? (b) How much electrical energy is consumed by the freezer during this hour? (c) How much wasted heat is delivered to the room in which the freezer sits?

20.11 •• A refrigerator has a coefficient of performance of 2.25, runs on an input of 135 W of electrical power, and keeps its inside compartment at 5°C. If you put a dozen 1.0 L plastic bottles of water at 31°C into this refrigerator, how long will it take for them to be cooled down to 5°C? (Ignore any heat that leaves the plastic.)

20.12 • A refrigerator has a coefficient of performance of 2.10. In each cycle it absorbs 3.10×10^4 J of heat from the cold reservoir. (a) How much mechanical energy is required each cycle to operate the refrigerator? (b) During each cycle, how much heat is discarded to the high-temperature reservoir?

Section 20.6 The Carnot Cycle

20.13 •• In a Carnot engine the hot reservoir is 72.0°C warmer than the cold reservoir. The engine's efficiency is 12.5%. What are the Kelvin temperatures of the two reservoirs?

20.14 • A Carnot engine is operated between two heat reservoirs at temperatures of 520 K and 300 K. (a) If the engine receives 6.45 kJ of heat energy from the reservoir at 520 K in each cycle, how many joules per cycle does it discard to the reservoir at 300 K? (b) How much mechanical work is performed by the engine during each cycle? (c) What is the thermal efficiency of the engine?

20.15 • A Carnot engine whose high-temperature reservoir is at 620 K takes in 550 J of heat at this temperature in each cycle and gives up 335 J to the low-temperature reservoir. (a) How much mechanical work does the engine perform during each cycle? What is (b) the temperature of the low-temperature reservoir; (c) the thermal efficiency of the cycle?

20.16 •• An ice-making machine operates in a Carnot cycle. It takes heat from water at 0.0°C and rejects heat to a room at 20.5°C. Suppose that 85.0 kg of water at 0.0°C are converted to ice at 0.0°C. (a) How much heat is discharged into the room? (b) How much energy must be supplied to the device?

20.17 • A Carnot engine performs 2.5×10^4 J of work in each cycle and has an efficiency of 66%. (a) How much heat does the engine extract from its heat source in each cycle? (b) If the engine exhausts heat at room temperature (20.0°C), what is the temperature of its heat source?

20.18 •• A certain brand of freezer is advertised to use 730 kW · h of energy per year. (a) Assuming the freezer operates for 5 hours each day, how much power does it require while operating? (b) If the freezer keeps its interior at −5.0°C in a 20.0°C room, what is its theoretical maximum performance coefficient? (c) What is the theoretical maximum amount of ice this freezer could make in an hour, starting with water at 20.0°C?

20.19 • A Carnot refrigerator is operated between two heat reservoirs at temperatures of 320 K and 270 K. (a) If in each cycle the refrigerator receives 415 J of heat energy from the reservoir at 270 K, how many joules of heat energy does it deliver to the reservoir at 320 K? (b) If the refrigerator completes 165 cycles each minute, what power input is required to operate it? (c) What is the coefficient of performance of the refrigerator?

20.20 •• A Carnot heat engine uses a hot reservoir consisting of a large amount of boiling water and a cold reservoir consisting of a large tub of ice and water. In 5 minutes of operation, the heat rejected by the engine melts 0.0400 kg of ice. During this time, how much work W is performed by the engine?

20.21 •• Carnot refrigerator A has a 16% higher coefficient of performance than Carnot refrigerator B. The temperature difference between the hot and cold reservoirs is 30% greater for B than for A. If the cold-reservoir temperature for refrigerator B is 180 K, what is the cold-reservoir temperature for refrigerator A?

Section 20.7 Entropy

20.22 • **CALC** You decide to take a nice hot bath but discover that your thoughtless roommate has used up most of the hot water. You fill the tub with 195 kg of 30.0°C water and attempt to warm it further by pouring in 5.00 kg of boiling water from the stove. (a) Is this a reversible or an irreversible process? Use physical reasoning to explain. (b) Calculate the final temperature of the bath water. (c) Calculate the net change in entropy of the system (bath water + boiling water), assuming no heat exchange with the air or the tub itself.

20.23 • A sophomore with nothing better to do adds heat to 0.350 kg of ice at 0.0°C until it is all melted. (a) What is the change in entropy of the water? (b) The source of heat is a very massive object at 25.0°C. What is the change in entropy of this object? (c) What is the total change in entropy of the water and the heat source?

20.24 •• **CALC** You make tea with 0.250 kg of 88.0°C water and let it cool to room temperature (20.0°C). (a) Calculate the entropy change of the water while it cools. (b) The cooling process is essentially isothermal for the air in your kitchen. Calculate the change in entropy of the air while the tea cools, assuming that all of the heat lost by the water goes into the air. What is the total entropy change of the system tea + air?

20.25 • Three moles of an ideal gas undergo a reversible isothermal compression at 17.0°C. During this compression, 1600 J of work is done on the gas. What is the change of entropy of the gas?

20.26 •• What is the change in entropy of 0.125 kg of helium gas at the normal boiling point of helium when it all condenses isothermally to 1.00 L of liquid helium? (*Hint:* See Table 17.4 in Section 17.5.)

20.27 • (a) Calculate the change in entropy when 1.00 kg of water at 100°C is vaporized and converted to steam at 100°C (see Table 17.4). (b) Compare your answer to the change in entropy when 1.00 kg of ice is melted at 0°C, calculated in Example 20.5 (Section 20.7). Is the change in entropy greater for melting or for vaporization? Interpret your answer using the idea that entropy is a measure of the randomness of a system.

Section 20.8 Microscopic Interpretation of Entropy

20.28 • **CALC** Two moles of an ideal gas occupy a volume V. The gas expands isothermally and reversibly to a volume $3V$. (a) Is the velocity distribution changed by the isothermal expansion? Explain. (b) Use Eq. (20.23) to calculate the change in entropy of the gas. (c) Use Eq. (20.18) to calculate the change in entropy of the gas. Compare this result to that obtained in part (b).

20.29 • **CALC** A lonely party balloon with a volume of 2.40 L and containing 0.100 mol of air is left behind to drift in the temporarily uninhabited and depressurized International Space Station. Sunlight coming through a porthole heats and explodes the balloon, causing the air in it to undergo a free expansion into the empty station, whose total volume is 425 m^3. Calculate the entropy change of the air during the expansion.

PROBLEMS

20.30 • CP As a budding mechanical engineer, you are called upon to design a Carnot engine that has 2.00 mol of a monatomic ideal gas as its working substance and operates from a high-temperature reservoir at 500°C. The engine is to lift a 15.0 kg weight 2.00 m per cycle, using 500 J of heat input. The gas in the engine chamber can have a minimum volume of 5.00 L during the cycle. (a) Draw a pV-diagram for this cycle. Show in your diagram where heat enters and leaves the gas. (b) What must be the temperature of the cold reservoir? (c) What is the thermal efficiency of the engine? (d) How much heat energy does this engine waste per cycle? (e) What is the maximum pressure that the gas chamber will have to withstand?

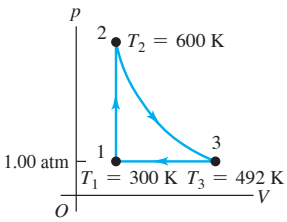
20.31 •• CP An ideal Carnot engine operates between 540°C and 170°C with a heat input of 200 J per cycle. (a) How much heat is delivered to the cold reservoir in each cycle? (b) What minimum number of cycles is necessary for the engine to lift a 430 kg rock through a height of 130 m?

20.32 •• BIO Entropy of Metabolism. An average sleeping person metabolizes at a rate of about 80 W by digesting food or burning fat. Typically, 20% of this energy goes into bodily functions, such as cell repair, pumping blood, and other uses of mechanical energy, while the rest goes to heat. Most people get rid of all this excess heat by transferring it (by conduction and the flow of blood) to the surface of the body, where it is radiated away. The normal internal temperature of the body (where the metabolism takes place) is 37°C, and the skin is typically 7°C cooler. By how much does the person's entropy change per second due to this heat transfer?

20.33 •• CP A certain heat engine operating on a Carnot cycle absorbs 410 J of heat per cycle at its hot reservoir at 135°C and has a thermal efficiency of 22.0%. (a) How much work does this engine do per cycle? (b) How much heat does the engine waste each cycle? (c) What is the temperature of the cold reservoir? (d) By how much does the engine change the entropy of the world each cycle? (e) What mass of water could this engine pump per cycle from a well 35.0 m deep?

20.34 • A heat engine takes 0.350 mol of a diatomic ideal gas around the cycle shown in the pV-diagram of **Fig. P20.34**. Process $1 \rightarrow 2$ is at constant volume, process $2 \rightarrow 3$ is adiabatic, and process $3 \rightarrow 1$ is at a constant pressure of 1.00 atm. The value of γ for this gas is 1.40. (a) Find the pressure and volume at points 1, 2, and 3. (b) Calculate Q, W, and ΔU for each of the three processes. (c) Find the net work done by the gas in the cycle. (d) Find the net heat flow into the engine in one cycle. (e) What is the thermal efficiency of the engine? How does this compare to the efficiency of a Carnot-cycle engine operating between the same minimum and maximum temperatures T_1 and T_2?

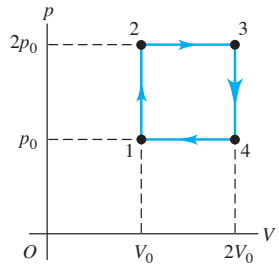
Figure P20.34



20.35 •• BIO Entropy Change from Digesting Fat. Digesting fat produces 9.3 food calories per gram of fat, and typically 80% of this energy goes to heat when metabolized. (One food calorie is 1000 calories and therefore equals 4186 J.) The body then moves all this heat to the surface by a combination of thermal conductivity and motion of the blood. The internal temperature of the body (where digestion occurs) is normally 37°C, and the surface is usually about 30°C. By how much do the digestion and metabolism of a 2.50 g pat of butter change your body's entropy? Does it increase or decrease?

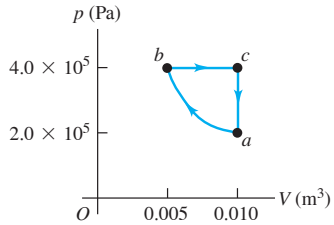
20.36 •• What is the thermal efficiency of an engine that takes n moles of diatomic ideal gas through the cycle $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$ shown in **Fig. P20.36**?

Figure P20.36



20.37 •• CALC You build a heat engine that takes 1.20 mol of an ideal diatomic gas through the cycle shown in **Fig. P20.37.** (a) Show that process ab is an isothermal compression. (b) During which process(es) of the cycle is heat absorbed by the gas? During which process(es) is heat rejected? How do you know? Calculate (c) the temperature at points a, b, and c; (d) the net heat exchanged with the surroundings and net work done by the engine in one cycle; (e) the thermal efficiency of the engine.

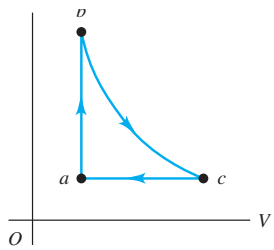
Figure P20.37



20.38 •• CALC One summer evening you fill a 0.3 L cup with ice and water to drink before going to bed. Estimate the volume of the ice and the volume of the water in the cup. Use the densities of ice and liquid water to find the mass of each. You set the cup on the kitchen counter, forget about it, and go to bed. The air in the kitchen is maintained at a constant 22°C by the air-conditioning system of your apartment. When you wake up in the morning, the cup contains liquid water at 22°C. (a) What is the entropy change of the ice water during the process in which it went from 0°C to liquid water at 22°C as you slept? (b) What is the entropy change of the air in the room due to the heat that flows from it into the contents of the cup? (c) What is the total entropy change of the air–ice water system due to the heat exchange that changes the phase and temperature of the ice water? Is the total entropy change positive or negative?

20.39 • CALC A heat engine operates using the cycle shown in **Fig. P20.39**. The working substance is 2.00 mol of helium gas, which reaches a maximum temperature of 327°C. Assume the helium can be treated as an ideal gas. Process bc is isothermal. The pressure in states a and c is 1.00×10^5 Pa, and the pressure in state b is 3.00×10^5 Pa. (a) How much heat enters the gas and how much leaves the gas each cycle? (b) How much work does the engine do each cycle, and what is its efficiency? (c) Compare this engine's efficiency with the maximum possible efficiency attainable with the hot and cold reservoirs used by this cycle.

Figure P20.39

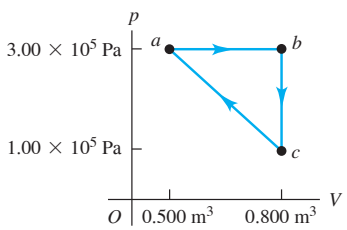


20.40 •• CP BIO **A Human Engine.** You decide to use your body as a Carnot heat engine. The operating gas is in a tube with one end in your mouth (where the temperature is 37.0°C) and the other end at the surface of your skin, at 30.0°C. (a) What is the maximum efficiency of such a heat engine? Would it be a very useful engine? (b) Suppose you want to use this human engine to lift a 2.20 kg box from the floor to a tabletop 1.40 m above the floor. How much must you increase the gravitational potential energy, and how much heat input is needed to accomplish this? (c) If your favorite candy bar has 350 food calories (1 food calorie = 4186 J) and 80% of the food energy goes into heat, how many of these candy bars must you eat to lift the box in this way?

20.41 ••• An experimental power plant at the Natural Energy Laboratory of Hawaii generates electricity from the temperature gradient of the ocean. The surface and deep-water temperatures are 27°C and 6°C, respectively. (a) What is the maximum theoretical efficiency of this power plant? (b) If the power plant is to produce 210 kW of power, at what rate must heat be extracted from the warm water? At what rate must heat be absorbed by the cold water? Assume the maximum theoretical efficiency. (c) The cold water that enters the plant leaves it at a temperature of 10°C. What must be the flow rate of cold water through the system? Give your answer in kg/h and in L/h.

20.42 • A monatomic ideal gas is taken around the cycle shown in **Fig. P20.42** in the direction shown in the figure. The path for process $c \rightarrow a$ is a straight line in the pV-diagram. (a) Calculate Q, W, and ΔU for each process $a \rightarrow b$, $b \rightarrow c$, and $c \rightarrow a$. (b) What are Q, W, and ΔU for one complete cycle? (c) What is the efficiency of the cycle?

Figure **P20.42**



20.43 • CALC A cylinder contains oxygen at a pressure of 2.00 atm. The volume is 4.00 L, and the temperature is 300 K. Assume that the oxygen may be treated as an ideal gas. The oxygen is carried through the following processes:

(i) Heated at constant pressure from the initial state (state 1) to state 2, which has $T = 450$ K.
(ii) Cooled at constant volume to 250 K (state 3).
(iii) Compressed at constant temperature to a volume of 4.00 L (state 4).
(iv) Heated at constant volume to 300 K, which takes the system back to state 1.

(a) Show these four processes in a pV-diagram, giving the numerical values of p and V in each of the four states. (b) Calculate Q and W for each of the four processes. (c) Calculate the net work done by the oxygen in the complete cycle. (d) What is the efficiency of this device as a heat engine? How does this compare to the efficiency of a Carnot-cycle engine operating between the same minimum and maximum temperatures of 250 K and 450 K?

20.44 ••• A typical coal-fired power plant generates 1000 MW of usable power at an overall thermal efficiency of 37%. (a) What is the rate of heat input to the plant? (b) The plant burns anthracite coal, which has a heat of combustion of 2.65×10^7 J/kg. How much coal does the plant use per day, if it operates continuously? (c) At what rate is heat ejected into the cool reservoir, which is the nearby river? (d) The river is at 18.0°C before it reaches the power plant and 18.4°C after it has received the plant's waste heat. Calculate the river's flow rate, in cubic meters per second. (e) By how much does the river's entropy increase each second?

20.45 •• A Carnot engine operates between two heat reservoirs at temperatures T_H and T_C. An inventor proposes to increase the efficiency by running one engine between T_H and an intermediate temperature T' and a second engine between T' and T_C, using as input the heat expelled by the first engine. Compute the efficiency of this composite system, and compare it to that of the original engine.

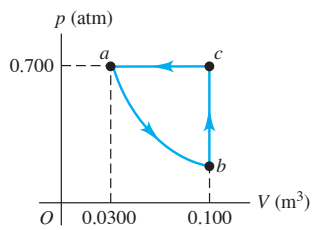
20.46 •• (a) Estimate the fuel consumption, in liters per 100 km, of a typical car. (b) The combustion of a liter of petrol releases 32 MJ of energy. If a typical car is driven at 60 km/h and the engine is turning at 3000 revolutions per minute, then based on your estimate of fuel efficiency, how many joules of energy are released by combustion during each cycle of the engine? (c) A typical car engine runs with 20% efficiency. Based on this figure, how many watts of power does the car supply when moving at a steady 60 km/h?

20.47 • **Automotive Thermodynamics.** A Skoda Superb has a six-cylinder Otto-cycle engine with compression ratio $r = 10.6$. The diameter of each cylinder, called the *bore* of the engine, is 82.5 mm. The distance that the piston moves during the compression in Fig. 20.5, called the *stroke* of the engine, is 86.4 mm. The initial pressure of the air–fuel mixture (at point a in Fig. 20.6) is 8.50×10^4 Pa, and the initial temperature is 300 K (the same as the outside air). Assume that 200 J of heat is added to each cylinder in each cycle by the burning petrol and that the gas has $C_V = 20.5$ J/mol · K and $\gamma = 1.40$. (a) Calculate the total work done in one cycle in each cylinder of the engine, and the heat released when the gas is cooled to the temperature of the outside air. (b) Calculate the volume of the air–fuel mixture at point a in the cycle. (c) Calculate the pressure, volume, and temperature of the gas at points b, c, and d in the cycle. In a pV-diagram, show the numerical values of p, V, and T for each of the four states. (d) Compare the efficiency of this engine with the efficiency of a Carnot-cycle engine operating between the same maximum and minimum temperatures.

20.48 •• (a) Estimate the size of your room. (b) One mole of an ideal gas at STP occupies a volume of 22.4 L. Using this figure, estimate the number of moles of air in your room. (c) How many molecules is that? (d) There are N^N ways to distribute N things onto N sites. As a rough approximation, assume there are as many states for air molecules as there are air molecules in your room. Compute the entropy of the air in your room.

20.49 •• The pV-diagram in **Fig. P20.49** shows the cycle for a refrigerator operating on 0.850 mol of H_2. Assume that the gas can be treated as ideal. Process ab is isothermal. Find the coefficient of performance of this refrigerator.

Figure **P20.49**



20.50 •• BIO **Human Entropy.** A person with skin of surface area 1.85 m^2 and temperature 30.0°C is resting in an insulated room where the ambient air temperature is 20.0°C. Assume that this person gets rid of excess heat by radiation only. By how much does the person change the entropy of the air in this room each second? (Recall that the room radiates back into the person and that the emissivity of the skin is 1.00.)

20.51 •• CALC An object of mass m_1, specific heat c_1, and temperature T_1 is placed in contact with a second object of mass m_2, specific heat c_2, and temperature $T_2 > T_1$. As a result, the temperature of the first object increases to T and the temperature of the second object decreases to T'. (a) Show that the entropy increase of the system is

$$\Delta S = m_1 c_1 \ln \frac{T}{T_1} + m_2 c_2 \ln \frac{T'}{T_2}$$

and show that energy conservation requires that

$$m_1 c_1 (T - T_1) = m_2 c_2 (T_2 - T')$$

(b) Show that the entropy change ΔS, considered as a function of T, is a *maximum* if $T = T'$, which is just the condition of thermodynamic equilibrium. (c) Discuss the result of part (b) in terms of the idea of entropy as a measure of randomness.

20.52 •• CALC To heat 1 cup of water $(250\ \text{cm}^3)$ to make coffee, you place an electric heating element in the cup. As the water temperature increases from 20°C to 78°C, the temperature of the heating element remains at a constant 120°C. Calculate the change in entropy of (a) the water; (b) the heating element; (c) the system of water and heating element. (Make the same assumption about the specific heat of water as in Example 20.10 in Section 20.7, and ignore the heat that flows into the ceramic coffee cup itself.) (d) Is this process reversible or irreversible? Explain.

20.53 •• DATA In your summer job with a venture capital firm, you are given funding requests from four inventors of heat engines. The inventors claim the following data for their operating prototypes:

	Prototype			
	A	B	C	D
T_C (°C), low-temperature reservoir	47	17	−33	37
T_H (°C), high-temperature reservoir	177	197	247	137
Claimed efficiency e (%)	21	35	56	20

(a) Based on the T_C and T_H values for each prototype, find the maximum possible efficiency for each. (b) Are any of the claimed efficiencies impossible? Explain. (c) For all prototypes with an efficiency that is possible, rank the prototypes in decreasing order of the ratio of claimed efficiency to maximum possible efficiency.

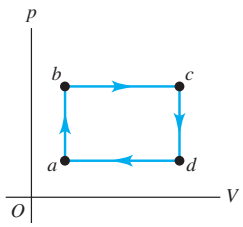
20.54 •• DATA For a refrigerator or air conditioner, the coefficient of performance K (often denoted as COP) is, as in Eq. (20.9), the ratio of cooling output $|Q_C|$ to the required electrical energy input $|W|$, both in joules. The coefficient of performance is also expressed as a ratio of powers,

$$K = \frac{|Q_C|/t}{|W|/t}$$

where $|Q_C|/t$ is the cooling power and $|W|/t$ is the electrical power input to the device, both in watts. (a) For a home air conditioner, K is generally determined for a 35°C outside temperature and an 27°C return air temperature. Calculate K for a Carnot device that operates between 35°C and 27°C. (b) You have an air conditioner with K = 3.2. Your home on average requires a total cooling output of $|Q_C| = 1.9 \times 10^{10}$ J per year. If electricity costs you 12.5 pence per kW · h, how much do you spend per year, on average, to operate your air conditioner? (Assume that the unit's COP accurately represents the operation of your air conditioner. A *seasonal coefficient of performance (SCOP)* is often used. The SCOP is calculated over a range of outside temperatures to get a more accurate seasonal average.) (c) You are considering replacing your air conditioner with a more efficient one with K = 4.3. Based on the value of K, how much would that save you on electricity costs in an average year?

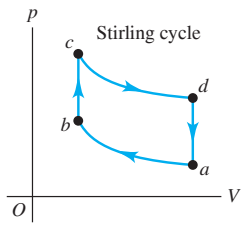
20.55 ••• DATA You are conducting experiments to study prototype heat engines. In one test, 4.00 mol of argon gas are taken around the cycle shown in **Fig. P20.55**. The pressure is low enough for the gas to be treated as ideal. You measure the gas temperature in states a, b, c, and d and find $T_a = 250.0$ K, $T_b = 300.0$ K, $T_c = 380.0$ K, and $T_d = 316.7$ K. (a) Calculate the efficiency e of the cycle. (b) Disappointed by the cycle's low efficiency, you consider doubling the number of moles of gas while keeping the pressure and volume the same. What would e be then? (c) You remember that the efficiency of a Carnot cycle increases if the temperature of the hot reservoir is increased. So, you return to using 4.00 mol of gas but double the volume in states c and d while keeping the pressures the same. The resulting temperatures in these states are $T_c = 760.0$ K and $T_d = 633.4$ K. T_a and T_b remain the same as in part (a). Calculate e for this cycle with the new T_c and T_d values. (d) Encouraged by the increase in efficiency, you raise T_c and T_d still further. But e doesn't increase very much; it seems to be approaching a limiting value. If $T_a = 250.0$ K and $T_b = 300.0$ K and you keep volumes V_a and V_b the same as in part (a), then $T_c/T_d = T_b/T_a$ and $T_c = 1.20T_d$. Derive an expression for e as a function of T_d for this cycle. What value does e approach as T_d becomes very large?

Figure **P20.55**



20.56 •• CP In an ideal Stirling engine, n moles of an ideal gas are isothermally compressed, pressurized at constant volume, allowed to expand isothermally, and then cooled at constant volume, as shown in **Fig. P20.56**.The expansion takes place at temperature T_H and the compression at temperature T_C. (a) Show that $p_c/p_d = p_b/p_a$. This quantity is called the compression ratio of the engine, or the CR. Determine the work done in terms of the CR (b) during the expansion phase and (c) during the compression phase. (d) An ideal Stirling engine uses 1 mole of helium as its working substance and has a CR of 10, temperature reservoirs at 100ºC and 20ºC, and a frequency of operation of 100 Hz. What is its power output?

Figure **P20.56**



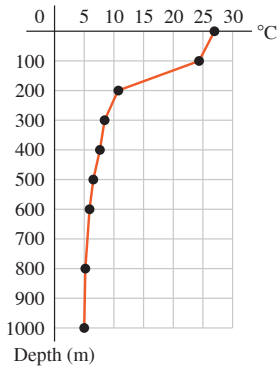
CHALLENGE PROBLEMS

20.57 ••• On planet Tralfamadore the cool atmosphere is rich in froklom molecules. Spherical life forms called sporons feed on these molecules and obtain energy through a Carnot cycle. When a sporon that has injested froklom gas has isothermally contracted to a trigger volume, it violently squeezes itself, reducing its volume by half. At this point, its interior achieves the combustion temperature at which froklom fuses exothermically to produce klombactic acid. At the point of combustion, a sporon has radius 8.0 cm and internal pressure 20.3 kPa. The sporon swells during the reaction period, maintaining constant temperature until its injested froklom is depleted. The sporon continues to swell rapidly until its internal temperature matches the ambient temperature of 123 K and its gauge pressure is zero. Then it contracts slowly, emitting klombactic acid while simultaneously injesting froklom from the atmosphere. When its radius shrinks by a factor of 3 it reaches the trigger volume, which starts another cycle. Aside from the sporon's skin membrane, all substances behave like ideal gases. Froklom gas has a ratio of heat capacities of 1.30. (a) What is the combustion temperature at which froklom fuses? (b) What is the ambient pressure on Tralfamadore? (c) If a sporon generates 60 kJ of work each hour, how much heat does it draw in per minute? (d) How much heat is rejected back to the atmosphere per minute?

20.58 ••• Consider a Diesel cycle that starts (at point a in Fig. 20.7) with air at temperature T_a. The air may be treated as an ideal gas. (a) If the temperature at point c is T_c, derive an expression for the efficiency of the cycle in terms of the compression ratio r. (b) What is the efficiency if $T_a = 300$ K, $T_c = 960$ K, $\gamma = 1.40$, and $r = 21.0$?

MCAT-STYLE PASSAGE PROBLEMS

Power from the Sea. *Ocean thermal energy conversion* is a process that uses the temperature difference between the warm surface water of tropical oceans and the cold deep-ocean water to run a heat engine. The graph shows a typical decrease of temperature with depth below the surface in tropical oceans. In the heat engine, the warmer surface water vaporizes a low-boiling-point fluid, such as ammonia. The heat of vaporization of ammonia is 260 cal/g at 27°C, the surface-water temperature. The vapor is used to turn a turbine and is then condensed back into a liquid by means of cold water



brought from deep below the surface through a large intake pipe. A power plant producing 10 MW of useful power would require a cold seawater flow rate of about 30,000 kg/s.

20.59 If the power plant uses a Carnot cycle and the desired theoretical efficiency is 6.5%, from what depth must cold water be brought? (a) 100 m; (b) 400 m; (c) 800 m; (d) deeper than 1000 m.

20.60 What is the change in entropy of the ammonia vaporized per second in the 10 MW power plant, assuming an ideal Carnot efficiency of 6.5%? (a) $+6 \times 10^6$ J/K per second; (b) $+5 \times 10^5$ J/K per second; (c) $+1 \times 10^5$ J/K per second; (d) 0.

20.61 Compare the entropy change of the warmer water to that of the colder water during one cycle of the heat engine, assuming an ideal Carnot cycle. (a) The entropy does not change during one cycle in either case. (b) The entropy of both increases, but the entropy of the colder water increases by more because its initial temperature is lower. (c) The entropy of the warmer water decreases by more than the entropy of the colder water increases, because some of the heat removed from the warmer water goes to the work done by the engine. (d) The entropy of the warmer water decreases by the same amount that the entropy of the colder water increases.

20.62 If the proposed plant is built and produces 10 MW but the rate at which waste heat is exhausted to the cold water is 165 MW, what is the plant's actual efficiency? (a) 5.7%; (b) 6.1%; (c) 6.5%; (d) 16.5%.

ANSWERS

Chapter Opening Question ?

(i) This is what a refrigerator does: It makes heat flow from the cold interior of the refrigerator to the warm outside. The second law of thermodynamics says that heat cannot *spontaneously* flow from a cold object to a hot one. A refrigerator has a motor that does work on the system to *force* the heat to flow in that way.

Key Example VARIATION Problems

VP20.1.1 (a) 6.89×10^4 J (b) 5.65×10^4 J
VP20.1.2 0.173
VP20.1.3 (a) 6.17×10^8 J (b) 1.21×10^8 J
VP20.1.4 (a) 3.96×10^8 J (b) 1.7×10^9 J (c) 0.23
VP20.4.1 (a) 0.210 (b) 6.32×10^4 J (c) 104°C
VP20.4.2 (a) 0.60
(b) $W_{ab} = 576$ J, $W_{bc} = 1250$ J, $W_{cd} = -231$ J, $W_{da} = -1250$ J
VP20.4.3 (a) 7.52 (b) 5.32×10^5 J
VP20.4.4 (a) $nRT_H \ln 2$ (b) $\frac{3}{2}nR(T_H - T_C)$
(c) $T_C/T_H = 1 - \frac{2}{3}\ln 2 = 0.538$, $e = \frac{2}{3}\ln 2 = 0.462$
VP20.10.1 (a) 6.55×10^2 J/K (b) 1.92×10^3 J/K (c) 2.43×10^3 J/K
VP20.10.2 (a) +45.7 J/K (b) +45.7 J/K
VP20.10.3 (a) zero (b) −150 J/K (c) +218 J/K (d) +68 J/K; increase
VP20.10.4 (a) $+1.22 \times 10^3$ J/K (b) -1.05×10^3 J/K
(c) $+1.7 \times 10^2$ J/K; increase

Bridging Problem

(a) 34.8°C
(b) +10 J/K